



PLS path modeling and evolutionary segmentation



Christian M. Ringle^{a,b,*}, Marko Sarstedt^{b,c}, Rainer Schlittgen^d, Charles R. Taylor^{e,**}

^a Hamburg University of Technology (TUHH), Institute for Human Resource Management and Organizations (HRMO), Schwarzenbergstr. 95 (D), 21073 Hamburg, Germany

^b University of Newcastle, Faculty of Business and Law, Callaghan NSW 2308, Australia

^c Ludwig-Maximilians-University (LMU) Munich, Institute for Market-based Management (IMM), Kaulbachstr. 45, 80539 Munich, Germany

^d University of Hamburg, Institute for Statistics and Econometrics, Von Melle-Park 5, 20146 Hamburg, Germany

^e Villanova University, School of Business, Bartley Hall Rm 2053, Marketing, 800 Lancaster Avenue, Villanova, PA 19085, USA

ARTICLE INFO

Article history:

Received 1 April 2011

Received in revised form 1 September 2011

Accepted 1 November 2011

Available online 3 March 2012

Keywords:

Partial least squares

Path modeling

Genetic algorithm

Segmentation

Heterogeneity

ABSTRACT

Applications of the partial least squares (PLS) path modeling approach—which have gained increasing dissemination in business research—usually build on the assumption that the data stem from a single population. However, in empirical applications, this assumption of homogeneity is unrealistic. Analyses on the aggregate data level ignore the existence of groups with substantial differences and more often than not result in misleading interpretations and false conclusions. This study introduces a genetic algorithm segmentation method for PLS path modeling (PLS-GAS) that accounts for the critical issue of unobserved heterogeneity in the path model's estimates of relations. The results from computational experiments allow a primary assessment to substantiate that PLS-GAS effectively uncovers unobserved heterogeneity. Significantly distinctive segment-specific path model estimates further foster the development of differentiated results that render more effective recommendations.

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1. Introduction

Researchers in marketing, management, and other related fields routinely use structural equation models to estimate relations between unobserved constructs and manifest variables. Although they traditionally rely on covariance structure analysis (e.g., Jöreskog, 1978) to estimate these models, the other main approach, partial least squares (PLS) path modeling (e.g., Lohmöller, 1989; Wold, 1974), has gained increasing dissemination in a variety of disciplines such as information systems (e.g., Rapp, Trainor, & Agnihotri, 2010; Ringle, Sarstedt, & Straub, 2012), and marketing (e.g., Hair, Sarstedt, Ringle, & Mena, forthcoming; Okazaki & Taylor, 2008). For instance, the article by Hair, Ringle & Sarstedt (2011) provides a general introduction to structural equation modeling using the PLS path modeling method.

When estimating path models, researchers usually assume that the analyzed data come from a single homogenous population. This assumption is often unrealistic (Jedidi, Jagpal, & DeSarbo, 1997). Specifically, the interpretation of the PLS path model estimations can be ambiguous, incomplete, and ineffective if the heterogeneous data involve significantly different segment-specific parameter estimates (Esposito Vinzi, Trinchera, Squillacciotti, & Tenenhaus, 2008; Rigdon,

Ringle, & Sarstedt, 2010). Therefore, researchers need to address the critical issue of heterogeneity by using effective analytical tools.

Researchers frequently rely on a priori information to partition the data into segments based on observable characteristics (e.g., socio demographic information). Alternatively, they apply traditional clustering procedures (e.g., k-means) to manifest variables or construct scores to identify distinct segments. However, neither approach provides satisfactory results for segmentation in structural equation modeling, especially as researchers frequently cannot observe heterogeneity nor do they know the true source (e.g., Jedidi et al., 1997; Sarstedt & Ringle, 2010). Consequently, methodological research provides several approaches for addressing the issue of uncovering and treating unobserved heterogeneity in PLS path modeling (Rigdon et al., 2010).

A review of these segmentation techniques discloses severe drawbacks limiting their applicability in many research situations (Sarstedt, 2008; Table 1). For example, finite mixture PLS (FIMIX-PLS; Hahn, Johnson, Herrmann, & Huber, 2002; Sarstedt, Becker, Ringle, & Schwaiger, 2011; Sarstedt & Ringle, 2010)—currently the most widely used approach in the field (Rigdon et al., 2010)—only considers heterogeneity in structural model estimates and imposes distributional assumptions on endogenous constructs, which runs contrary to PLS path modeling's nonparametric character. As an extension of the PLS typological path modeling (PLS-TPM; Squillacciotti, 2010) procedure, the more recently proposed response-based unit segmentation in PLS path modeling (REBUS-PLS; Esposito Vinzi et al., 2008), a hill-climbing method, overcomes some limitations of FIMIX-PLS (Esposito Vinzi, Ringle, Squillacciotti, & Trinchera, 2007). However, REBUS-PLS suffers from several drawbacks. For example, REBUS-PLS determines the

* Correspondence to: C.M. Ringle, Hamburg University of Technology (TUHH), Institute for Human Resource Management and Organizations (HRMO), Schwarzenbergstr. 95 (D), 21073 Hamburg, Germany.

** Corresponding author.

E-mail addresses: hrm@tuhh.de (C.M. Ringle), sarstedt@bwl.lmu.de (M. Sarstedt), rainer.schlittgen@uni-hamburg.de (R. Schlittgen), raymond.taylor@villanova.edu (C.R. Taylor).

Table 1
Segmentation techniques for PLS path modeling.

	FIMIX-PLS	REBUS-PLS	PLS-GAS
Consideration of heterogeneity in...			
...the structural model	X	X	X
...reflective measurement models		X	X
...formative measurement models			X
The method does not rely on distributional assumptions		X	X
Potential escape of local optimum solutions			X
Availability of criteria for deciding on a suitable number of segments	X		

starting partition and the number of segments by using a hierarchical clustering method on structural model residuals from a model estimation on the aggregate data level. This initial step is problematic for at least two reasons. First, the residuals from a model estimation on the aggregate data level disregard the statistical model under consideration and therefore provide little guidance for determining the number of segments and a suitable initial segmentation solution. Second, when relying on a single, nonrandom starting partition, hill-climbing approaches generally perform very poorly as they cannot explore a sufficiently large fraction of the overall search space (e.g., Steinley, 2003). This problem worsens because hill-climbing approaches tend to prematurely converge in local optima.

In light of the considerable biases from disregarding heterogeneity (e.g., Jedidi et al., 1997), recent calls exist to routinely evaluate PLS path modeling results for unobserved heterogeneity (e.g., Hair et al., forthcoming; Henseler, Ringle, & Sinkovics, 2009). Against this background, this paper presents an evolutionary segmentation approach that overcomes critical deficiencies in the available methods. This method combines a genetic algorithm with a deterministic hill-climbing approach within a PLS path modeling framework. The partial least squares genetic algorithm segmentation (PLS-GAS) approach uses directed random searches to disclose an optimum solution in the complex search space that underlies segmentation tasks. The PLS-GAS method does not rely on distributional assumptions and, thus, matches the PLS path modeling method's nonparametric character. Moreover, PLS-GAS potentially allows uncovering heterogeneity in both reflective and formative measurement models. Thereby, the method is generally applicable to segmentation tasks in PLS path models commonly encountered in business research studies (Diamantopoulos, Riefler, & Roth, 2008; Gudergan, Ringle, Wende, & Will, 2008).

To evaluate the new approach's capabilities, this study presents several computational experiments with artificial datasets and contrasts PLS-GAS's performance with that of FIMIX-PLS and REBUS-PLS. The results substantiate the effectiveness of PLS-GAS across a wide range of data constellations, as well as the favorable performance of PLS-GAS compared to alternative segmentation techniques for PLS path modeling.

2. Genetic algorithm segmentation in partial least squares path modeling

2.1. Method

Segmentation tasks in PLS path modeling require the estimation of distinct segment-specific path models based on an assignment of data to K segments. In general terms, the identification of an optimal solution requires a segmentation procedure to consider all possible combinatorial data assignment solutions. However, the number of possible clustering solutions increases exponentially with higher numbers of observations and segments (Cowgill, Harvey, & Watson, 1999). Even small and mid-sized problems become extremely complex (e.g., $N=300$ observations and $K=2$ segments entail the possibility to form approximately 10^{90} different data partitions, an

amount that is much higher than the number of atoms in the known universe), making an exhaustive search by comparing all possible data assignment solutions infeasible.

To handle this complexity, this new approach combines the PLS path modeling algorithm with a genetic algorithm for data segmentation (Maulik & Bandyopadhyay, 2000). Well known for their capability to cope with complex multimodal search spaces (Michalewicz, 1996), genetic algorithms have become an accepted tool for carrying out complex segmentation tasks in marketing research (e.g., Fish, Johnson, Dorsey, & Blodgett, 2004). Specifically, as a genetic search/hill-climbing hybrid (Cowgill et al., 1999), PLS-GAS initially uses a non-deterministic genetic algorithm for data segmentation to uncover a solution that is at least close to the global optimum by using a guided search to explore the search space (first stage). Because genetic algorithm segmentation results do not necessarily provide a global optimum solution, the best segmentation result becomes the starting partition for a deterministic hill-climbing approach in order to locally improve, if possible, the genetic algorithm segmentation outcome and to ultimately determine the best partition (second stage).

2.2. First stage

In the first stage, PLS-GAS uses a genetic algorithm for data segmentation to generate a suitable solution for the segmentation task. Specifically, PLS-GAS uses a set of potential solutions and initiates a process analogous to the evolution process in nature. This process comprises the selection, reproduction, recombination, mutation, and replacement stages and continues until the process satisfies a termination criterion (Maulik & Bandyopadhyay, 2000).

The algorithm starts by randomly generating a population with a large number (e.g., 1000 or more) of I individuals ($i=1, \dots, I$). Every individual embodies a potential solution for partitioning the number of N observations ($n=1, \dots, N$) into a prespecified number of K segments ($k=1, \dots, K$). The individual's chromosome encodes this partitioning, whereby each chromosome's position (gene) is an allocation of the respective n -th observation to one of the K segments. Thus, an individual's chromosome represents a potential PLS path modeling segmentation solution. In the following simplified example with $N=7$ and $K=3$, the chromosome 1223311 represents a solution where the first and the last two observations belong to segment one, but the remaining observations belong to segments two and three, respectively.

To assess each solution's quality, PLS-GAS uses the estimates of each individual's segment-specific PLS path modeling result to compute an objective function value. This value corresponds to the path model's unexplained variance per segment. Specifically, PLS-GAS uses the following fitness criterion F for each individual i , where σ_{nmk}^2 symbolizes the residual term of the m th manifest variable with regard to the n th observation in the k th segment, and ζ_{nlk}^2 represents the structural model residual of the l th endogenous construct for the n th observation in the k th segment.

$$F_i = \sum_{nmk} \sigma_{nmk}^2 + \sum_{nlk} \zeta_{nlk}^2 \quad (1)$$

The PLS-GAS method aims to identify the individual (i.e., a data assignment solution of N observations into K segments) with the minimum value for F_i which thus represents an (at least local) optimum solution. This objective function fits the prediction-oriented character of the PLS path modeling method that aims at minimizing the latent endogenous variables' unexplained variance (Wold, 1974). At the same time, the objective function applies universally, regardless of whether the specifications of the measurement models are formative or reflective (e.g., Gudergan et al., 2008).

After initialization, PLS-GAS randomly selects individuals to populate the next generation. This selection process is a function of

each individual's F_i value that determines its rank r_i and rank fitness $I + 1 - r_i$ in the population. Individuals with a higher rank fitness (i.e., lower F_i value) have a greater probability of selection for the next generation, given by

$$p_i = (I + 1 - r_i) / (I(I + 1) / 2). \quad (2)$$

Selected individuals (parents) then randomly undergo one of three alternative genetic operations: recombination, mutation, or reproduction (Maulik & Bandyopadhyay, 2000; Michalewicz, 1996). These operations vary the chromosomes from one generation to the next (see the Appendix A for a description of each operation's conditions). Fig. 1 illustrates the working principle of PLS-GAS' first stage.

The recombination operation combines two parents to produce a new individual (offspring) by means of a single-point crossover of their chromosomes. Accordingly, the crossover operator randomly selects a crossover point within the parents' chromosomes, and then interchanges the two parent chromosomes at this point to produce two offspring (i.e., two new data assignment solutions). In the case of mutation, the operator randomly alters one or more genes in the selected parent's chromosome. Lastly, in the case of reproduction, an unmodified copy of the selected parent becomes a member of the next generation.

Regardless of these genetic operations, PLS-GAS always retains the best individual with the highest rank fitness for the next generation (elitist selection; Cowgill et al., 1999). This way, the algorithm populates a new generation with individuals and uses validity tests to ensure that each individual embodies a valid solution in terms of segment-specific sample size requirements.

By repeating this generational process 1000 or more times, PLS-GAS gradually improves the quality of the segmentation solution. As a terminating condition, PLS-GAS stops after populating a fixed number of generations.

2.3. Second stage

An important aspect of the genetic algorithm's stochastic operators is their capability to escape local optima in order to identify individuals with lower F_i values. However, because of the genetic operators' stochastic nature, the first stage does not necessarily indicate a global optimum solution. As a consequence, researchers

routinely combine heuristic approaches, such as genetic algorithms, with local search procedures (e.g., Cowgill et al., 1999). Similarly, PLS-GAS selects the best solution from the first stage as the starting partition for the second stage that performs local improvement by applying a deterministic hill-climbing approach. Using a procedure analogous to the iterative refinement approach that REBUS-PLS employs (Esposito Vinzi et al., 2008), local improvement in the second stage of PLS-GAS investigates whether an observation's alternative segment assignment further reduces the objective function value. Specifically, the procedure selects the first observation, gradually relocates this observation to each alternative segment, and recomputes the objective function F_i for each of the observation's assignments to one of the alternative segments. The algorithm then reallocates this observation to the segment corresponding to the minimum F_i value. If the observation remains in its original segment and, thus, the data assignment solution does not change, the analysis continues with the next observation. Otherwise, the procedure updates the segment-specific PLS path model estimates and starts anew. This hill-climbing process repeats until no reallocation occurs in a single loop, or the algorithm reaches the maximum number of prespecified iterations. The PLS-GAS method designates the final allocation of observations associated with the minimum objective function value as the final best segmentation result.

3. Computational experiments

3.1. Design, data generation, and analysis

The initial assessment of PLS-GAS by means of computational experiments builds on a path model with two exogenous constructs (ξ_1 and ξ_2), each with a relation to one endogenous construct (η_1) by means of two relations with coefficients ω_{11} and ω_{21} (Fig. 2); x and y represent sets of reflective manifest variables linked to their corresponding exogenous (endogenous) constructs through loadings π (γ).

The three measurement models have an identical number of manifest variables and all loadings in the reflective measurement models are at the same level. This computational experiment considers a two-segment model, in which the first segment has a higher ω_{11} and a lower ω_{21} value, and vice versa for the second segment. Because the coefficients' absolute values do not matter, but their distinctiveness

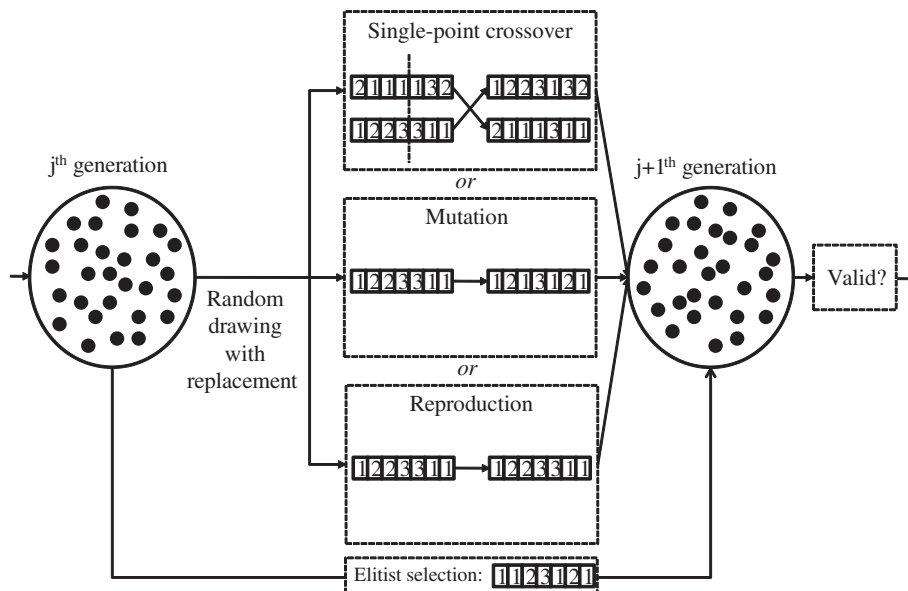


Fig. 1. Working principle of PLS-GAS's first stage.

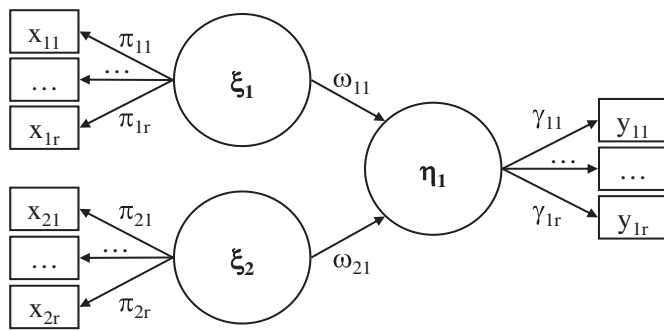


Fig. 2. PLS path model.

does, the computational experiment focuses on the alternative levels of differences in the structural model.

The identification of relevant data characteristics that potentially influence the performance of PLS-GAS builds on a review of the previous simulation studies on PLS path modeling (e.g., Reinartz, Haenlein, & Henseler, 2009) and segmentation contexts (e.g., Sarstedt et al., 2011). To provide a comprehensive evaluation of PLS-GAS, the computational experiment considers a multitude of factors and levels, which extends the simulation setups in prior research studies (e.g., Chin & Newsted, 1999; Esposito Vinzi et al., 2007). Specifically, this study considers the following factors: (1) the number of observations [100, 200, and 600], (2) the number of manifest variables in the measurement models [3, 6, and 12], (3) the relative sizes of the two segments [50%/50%, 70%/30%, and 90%/10%], (4) the mean separation between the segment-specific path coefficients [high, high to medium, medium to low, and low], and (5) the error variance of the endogenous construct and the manifest variables in relation to their respective total variance [0%, 5%, 10%, 15%, 20%, 25%, and 30%]. The design is factorial and with ten datasets per cell, this study includes a total of $3^3 \cdot 4 \cdot 7 \cdot 10 = 7560$ datasets.

In accordance with Sarstedt et al.'s (2011) data generation procedure, a GAUSS 9.0 program generates the datasets for the computational experiments. As preliminary evaluations show, the procedure provides sets of data that meet the expected values for the prespecified factor-level combinations. In addition, the resulting PLS path modeling parameters estimates' standard deviation from the ten data generation runs for a specific factor level combination is always smaller than 0.10. Hence, the data generation process yields reliable and valid parameter estimates.

In addition to a stand-alone assessment of PLS-GAS, for performance benchmarks, the analysis also comprises a comparison of the approach's performance with that of alternative data segmentation techniques for PLS path modeling. Specifically, this study considers FIMIX-PLS and REBUS-PLS, the two most salient and frequently applied methods for uncovering unobserved heterogeneity in PLS path modeling (e.g., Esposito Vinzi et al., 2008; Hahn et al., 2002; Rigdon et al., 2010; Sarstedt, 2008; Sarstedt & Ringle, 2010). This study's computational experiments draw on the following computer software tools: SmartPLS 2.0 (Ringle, Wende, & Will, 2005; FIMIX-PLS), the R module plspm (Sánchez & Trinchera, 2010; REBUS-PLS), and a GAUSS 9.0 implementation of PLS-GAS. The PLS-GAS method requires some decisions regarding the parameter settings for the first stage (genetic algorithm) in order to extensively analyze the complex search space: (1) the number of individuals in each population (i.e., $I = 5 \cdot N \cdot K$); (2) the probability of crossover (i.e., $p_c = 0.6$); (3) the probability of mutation (i.e., $p_m = 0.1$); (4) the probability of reproduction (i.e., $p_r = 0.3$); and (5) the number of generations in Stage 1 (i.e., $J = 5 \cdot N \cdot K$). The second stage stops after a maximum of 30 iterations or if, throughout a single iteration, no reassignment of an observation occurs. The FIMIX-PLS method only requires the selection of two parameters. In order to ensure that the algorithm does not converge

prematurely, FIMIX-PLS stops when the improvement of the log-likelihood value is smaller than 10^{-15} , or the iterations attain the maximum number of 30,000 (Sarstedt et al., 2011). Further, the study follows Sánchez and Trinchera's (2010) recommendations for REBUS-PLS by using a threshold where less than 0.05% of the observations change segments from one iteration to the other as the stopping rule, with a maximum of 100 iterations. Owing to FIMIX-PLS' possible convergence in local optimum solutions, each analysis draws on 30 replications.

3.2. Analysis of results

The first results of the analysis address the potential of PLS-GAS to meet the global optimum solution when no error variance affects the data and to control for whether the new method operates accurately under the best possible conditions for data segmentation purposes. The results of this analysis show that PLS-GAS provides a 100% correct assignment of observations in all data constellations that the error variance does not affect. Further analyses show that neither the number of observations nor the number of manifest variables per measurement model affects the results of PLS-GAS significantly, because they meet the given factor-level specifications. In addition, the relative segment sizes do not significantly affect the outcomes of PLS-GAS.

The results of the analyses also show that only two factors—the decreased distinctiveness of the segment-specific path coefficients and the increased level of error variance—reduce the data's separability and, thus, negatively affect the performance of PLS-GAS. Fig. 3(a) and (b) illustrate the development of the average differences of path coefficients across groups and varying levels of error variance for the artificially generated sets of data and the corresponding PLS-GAS results for 100 observations per segment and six manifest variables per construct. These results typify all the constellations in the analytical design as the sample size, the numbers of manifest variables per measurement model, and the relative segment sizes do not significantly influence the performance of PLS-GAS. The analyses show that the solutions from PLS-GAS entail segment-specific path coefficient differences (Fig. 3(b)) that consistently match the levels expected from the artificial datasets (Fig. 3(a)). Moreover, Fig. 3(e) and (f) illustrate the endogenous latent variable's average R^2 values regarding the artificial data as well as the corresponding results for the PLS-GAS analyses. Generally, all average R^2 values meet the expectations regarding the a priori generated sets of data. This finding also holds for increased levels of error variance and small differences in segment-specific path coefficients. Even though these two factors significantly increase the fuzziness of the data, PLS-GAS always performs well across all factor-level combinations. More detailed analyses show that the combination of the genetic algorithm and the hills-climbing approach proves valuable, as the second stage yields a different, and therefore improved, solution in 84.85% of the simulation runs.

Applying FIMIX-PLS and REBUS-PLS in the next step of the analysis shows that neither approach can reliably retain the prespecified segment structure. More precisely, even though FIMIX-PLS shows differences in the path coefficients between the two segments, their levels are independent from the prespecified levels and show little variation with regard to changes in error variance (Fig. 3(c)). While this approach achieves high levels of average R^2 values (Fig. 3(g)), the results do not match the expected patterns (Fig. 3(e)). Even worse, REBUS-PLS retains only marginal differences in the path coefficients, regardless of the prespecified difference levels and error variance (Fig. 3(d)). Simultaneously, and as expected, the average R^2 values decrease with increasing error variance. However, the average R^2 values are higher for lower levels of mean separation between the segment-specific path coefficients (Fig. 3(h)). Overall, limits clearly exist to the capabilities of FIMIX-PLS and REBUS-PLS to identify a prespecified segment structure. Conversely, the results of PLS-GAS closely match those expected from the artificially generated

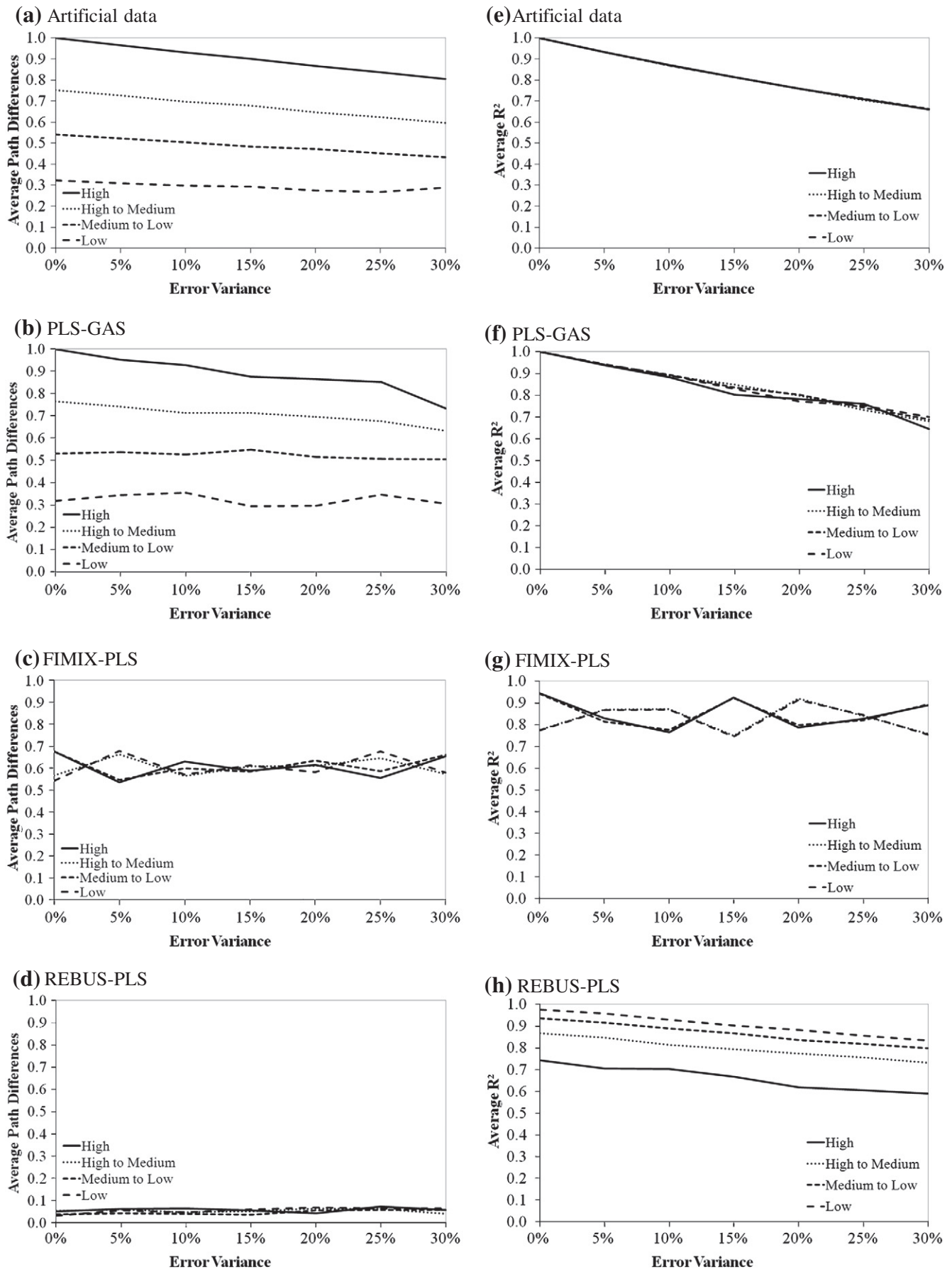


Fig. 3. Results (100 observations per segment and 6 manifest variables per construct).

datasets, supporting the procedure's capability to effectively uncover segments in heterogeneous data structures.

3.3. Additional analyses

To further evaluate the capabilities of PLS-GAS, additional analyses extend the initial model setup by evaluating the performance of PLS-GAS with non-normal data. The non-normal data generation builds on independent normal random variates; then, exponentiation transforms these data into lognormal random variates. Another extension of the study considers a three-, four-, and five-segment solution by incorporating an additional exogenous construct per additional segment. The parameter prespecification to generate the artificial data for these instances analogizes well to the two-segment example, where one specific segment has a very strong relation in the structural model, while all the other paths remain at lower levels.

The results show that the findings regarding the normal data scenario also hold true for situations with non-normal data. As with normal data, the average path coefficient differences and R^2 values based on the results from PLS-GAS all have similar levels if compared to the artificial data. The latter is an important finding with respect to PLS path modeling applications, because the majority of the data collected in behavioral research do not follow multivariate normal distributions.

A further examination of PLS-GAS by means of three, four, and five segments shows analogous results. The PLS-GAS method also provides a 100% correct rate of assignment of higher segment numbers in both normal and non-normal data constellations if error variance does not affect the data. The findings regarding the effects of other factor levels with respect to two segments also apply to constellations with higher numbers of segments and an increased degree of model complexity, showing that only the mean separation between segment-specific path coefficients and the error variance have a significant, albeit limited, effect on PLS-GAS's results.

4. Summary and conclusion

When using structural equation modeling, disregarding the existence of groups of data with substantially different parameter estimates can substantially bias the results. Existing approaches for dealing with such heterogeneous data structures suffer from several limitations that relate to, for example, their ability to explore a sufficiently large fraction of the overall search space or their capability to cope with formative measurement models. By introducing and assessing a novel evolutionary segmentation approach, this paper contributes to the need for effective means to treat unobserved heterogeneity in PLS path modeling (Rigdon et al., 2010).

The proposed genetic/hill-climbing hybrid approach ensures an extensive search of the space of possible data assignment solutions, a potential escape from local optima, and a solution that is at least close to the global optimum. The computational experiments in this study, which considerably extend the simulation setups that prior research studies present (e.g., Chin & Newsted, 1999; Esposito Vinzi et al., 2007), provide a comprehensive evaluation of the PLS-GAS method. Across a wide range of data constellations, including all combinations of systematically changed numbers of observations, numbers of manifest variables, relative segment sizes, prespecified numbers of segments, and regardless of the normality or non-normality of the data, the PLS-GAS results closely match those expected from the artificial datasets. Especially, the robustness of PLS-GAS against highly unbalanced segments is advantageous, considering the frequent use of such setups in empirical studies and the strong tendency of distance measure-based PLS segmentation methods, such as PLS-TPM and REBUS-PLS, to form equally sized data groups (Esposito Vinzi et al., 2007). The analysis of results also shows that the segments' decreased distinctiveness does not

necessarily have a pronounced negative impact on the capabilities of PLS-GAS. Nevertheless, even though the declining distinctiveness of segment-specific path coefficients reduces the correct rate of data assignment, the final segment-specific results consistently meet the artificially generated datasets' prespecified levels of path differences and R^2 values.

Despite the apparent usefulness of PLS-GAS, the approach and this study have certain limitations. Similar to PLS-TPM and REBUS-PLS, the PLS-GAS method fails to provide a clear indication of how to identify an appropriate number of segments. Usually, researchers merely revert to proxies, such as the solution's interpretability or acceptable R^2 values. Alternatively, researchers can use FIMIX-PLS on the data to guide this decision. Unlike competing procedures, FIMIX-PLS allows for the computation of information and classification criteria whose effectiveness in selecting the correct number of segments Sarstedt et al. (2011) recently evaluated.

This study focuses on evaluating the performance of the PLS-GAS with reflective measurement model setups. Given the ongoing debate on the appropriateness of formative measures (Diamantopoulos et al., 2008) and the ability of PLS path modeling to readily incorporate these, future research should evaluate the performance of PLS-GAS in this regard. Another avenue of further research refers to the PLS-GAS method's parameter settings. This study's choice of parameters ensures that PLS-GAS potentially covers a large fraction of the overall search space. However, simulation studies must indicate the improved settings that allow high quality solutions with less exhaustive and, thus, more efficient searches. Finally, evaluating the effectiveness of alternative objective functions that, for example, make use of communality and redundancy measures is another area for future research. These areas of future research improve researchers' knowledge of PLS-GAS and provide specific recommendations that foster the applicability of the method in research practice.

In conclusion, PLS-GAS offers advantageous capabilities for a wide range of data situations that researchers and practitioners typically encounter in studies. The method is a particularly valuable approach to addressing the critical issue of heterogeneous data structures in PLS path modeling. The PLS-GAS method effectively assists researchers and practitioners in uncovering relevant segments with their characteristic PLS path modeling results. This kind of analytical outcome is important for creating segment-specific interpretations and for pointing out better conclusions for decision makers.

Acknowledgments

Prior versions of this paper were presented at the 2010 Global Marketing Conference in Tokyo (Best Paper Award in the Advancing Research Methods track), Japan, and at the 2009 Annual Conference of the ANZMAC in Melbourne, Australia. The authors thank Manfred Schwaiger (Ludwig-Maximilians-University Munich), Edward E. Rigdon (Georgia State University) and the anonymous reviewers for their helpful remarks on previous versions of the manuscript.

Appendix A. Genetic operators

To randomly select the genetic operators, PLS-GAS draws a random number α ($0 < \alpha < 1$): (a) recombination, if $\alpha \leq p_c$; (b) mutation, if $p_c < \alpha \leq (p_c + p_m)$; (c) reproduction, if $\alpha > (p_c + p_m)$, where p_c is the probability for recombination, p_m the probability for mutation and p_r the probability for the reproduction with $p_c + p_m + p_r = 1$. In the case of recombination, a random integer h between 1 and $N - (1 < h \leq (N - 1))$ determines the crossover point where both chromosomes are divided. The first offspring inherits the first to h th gene from the first chromosome and the $(h + 1)$ th to the N th gene from the second chromosome, while the opposite holds for the second offspring. In the case of mutation, each observation's assignment to a specific segment is subject to mutation (i.e., gene transformation through the

observation's random reassignment to an alternative segment) with a prespecified probability in the parameter settings (i.e., $p_t = 0.05$). Finally, in the case of reproduction, an exact copy of an individual becomes a member of the next generation.

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